



Flow development in vertical annular flow

A. Wolf^a, S. Jayanti^{b,*}, G. F. Hewitt^a

^a*Department of Chemical Engineering, Imperial College, London SW7 2BY, UK*

^b*Department of Chemical Engineering, Indian Institute of Technology Madras, Chennai 600 036, India*

Received 19 April 2000; accepted 21 November 2000

Abstract

Air–water upward annular flow experiments were conducted in a 10.8 m long tube of 31.8 mm internal diameter. Local values of the pressure gradient, film thickness, wall shear stress, film flow rate, disturbance wave velocity and frequency were measured over a range of gas and liquid flow rates. The results show that rapid changes in the film flow parameters occur within the first 50 tube diameters or so from the inlet. Entrained fraction and the film thickness appear to be the slowest to respond and these may take 100 to 300 tube diameters to develop fully. © 2001 Elsevier Science Ltd. All rights reserved.

Keywords: Gas liquid flow; Annular flow; Flow development; Interfacial waves; Film flow; Wall shear stress

1. Introduction

In vertical gas–liquid annular flow, part of the liquid phase flows in the form of a thin liquid film and the rest is dispersed into the gas core in the form of fine droplets. The gas–liquid film interface is usually covered with two types of waves: ripple waves, which are small in amplitude and are coherent over short axial distances only; and disturbance waves, which have an amplitude of about five times the mean film thickness and are coherent over long (several meters) axial distances (Hewitt & Hall-Taylor, 1970). Annular flow occurs over a wide range of flow conditions and is the predominant flow regime in many steam generators and heat exchangers. The flow regime has been very extensively studied and detailed models have been proposed to calculate the pressure gradient and the mean film thickness and other characteristics of the flow regime (Asali, Leman, & Hanratty, 1985; Owen & Hewitt, 1987; Govan, 1990). Several studies (see for example, Hall-Taylor, 1966; Gill, Hewitt, & Roberts, 1969; Hewitt & Nicholls, 1969; Tomida & Okazaki, 1974; Whalley, Hewitt, Azzopardi, & Pshyk, 1977; Martin, 1983; Wilkes, Azzopardi, & Thompson, 1983; Azzopardi, 1986; Sekoguchi & Takeishi, 1989; Wolf, Jayanti, & Hewitt, 1996; Sekoguchi & Mori, 1997)

have also been made of the nature and characteristics of disturbance waves on the gas–liquid interface.

One of the important areas yet to be investigated thoroughly in annular flow is the evolution of the flow characteristics leading to the fully developed flow under hydrodynamic equilibrium. Brown, Jensen, and Whalley (1975) reported measurements of pressure gradient and entrainment obtained in a 31.8 mm internal diameter tube using two different liquid inlet devices, namely, a porous wall inlet and an axial jet injector, at distances of 140, 250, 310, 420 and 560 tube diameters from the liquid inlet. They found that differences between the two inlet devices persisted even at 550 pipe diameters indicating that the flow was not yet fully developed. They also reported measurements of wave velocity, wave frequency and pressure gradient in steam–water flow in a tube of 9.61 mm internal diameter at distances of 20, 50, 82, 114, 146, 177 and 209 diameters from the inlet. They found that the wave velocity and the dominant wave frequency did not change much with distance while the pressure gradient changed significantly which they attributed to the change in the test section pressure from 3.2 bar at inlet to 2.52 bar at outlet. The wave spectrum also took a fairly long distance to develop. They also investigated the effect of using different heated lengths to achieve the same quality and found that the difference, if any, between the two heated lengths persisted only for about 50 tube diameters even for pressure gradient and the wave spectrum.

* Corresponding author. Tel.: + 91-44-4458227; fax: + 91-44 2350509.

E-mail address: sjayanti@iitm.ac.in (S. Jayanti).

There is thus conflicting evidence for the length required for annular flow to become fully developed. It appears that some parameters such as wave spectrum and entrainment fraction take a fairly long distance to develop while other parameters such as the wave velocity and pressure gradient will develop much more quickly. The wave spectrum, i.e., the range of dominant frequencies, appears to be a strong function only of the length of development while the entrainment depends very strongly on the inlet device. Although these trends are reflective of the underlying phenomena occurring in annular flow, the inter-relation between them cannot be established because all the parameters are not measured in the same experiments. For example, the entrainment fraction was measured only in the air–water experiments of Brown et al. (1975) while the wave spectrum and velocity were measured in steam–water experiments in a much smaller tube diameter. The true picture would emerge only if all the flow parameters were measured under identical conditions. In view of this, a programme of work has been undertaken at Imperial College, London to measure all the important parameters of annular flow, namely, film thickness, wave characteristics, film flow rate, wall shear stress and pressure gradient, at several distances from the inlet so as to study the film development/evolution with distance from the inlet. It was also expected that the simultaneous measurement of the wall shear stress and the film thickness would enable one to gain better insight into the hydrodynamics of the film flow. The results of these experiments are discussed in this paper.

2. Experimental details

Air–water upward annular flow experiments were conducted at Imperial College, London in a 10.8 m long copper pipe of 31.8 mm internal diameter. This was a shorter version of the original LOTUS facility at Harwell Laboratory, UKAEA, used, among others, by Brown et al. (1975) for their air–water experiments. Water was introduced into the test section through a porous walled tube located 0.3 m from the air inlet at the bottom of the pipe. Air was taken from the compressed air supply system of Imperial College which delivered air at nearly constant pressure. The water flow rate through the inlet was measured using a bank of rotameters. The air flow rate was measured using an orifice plate. Details of the measurement of the other parameters are as follows:

- The pressure drop, over a distance of 0.83 m of the test section, was measured using an electronic differential pressure transducer with a maximum range of 350 mbar. A personal computer (PC) with an analogue-to-digital (A/D) card was used to record the output from the pressure transducer. Its reading was averaged over a time interval of 30 s. The set-up for measuring the pressure drop was validated by measuring the pressure drop in single phase flow over a distance 3.87 m. In the experiments, the pressure gradient was measured at distances of 0.67, 1.59, 3.2 and 9.74 m downstream of the water inlet.
- The film flow rate was measured directly by sucking off the film. A brass body containing a 7 cm long brass sinter was manufactured and was flanged in such a way that it could be inserted at different axial positions on the pipe. The outlet of the device was connected to a cyclone separator by means of a PVC hose. The liquid volume collected in the cyclone over a given time interval was noted and is used to calculate the liquid film flow rate. The device was calibrated by measuring the film flow rate under ripple flow conditions where, because there are no disturbance waves and hence no entrainment of droplets, the film flow rate is the same as the inlet liquid flow rate. For high film flow rates, a 7 cm long sinter was found to be too short to remove the film completely, and a 20 cm long sinter was used in these cases. In the experiments, the liquid film flow rate was measured at distances of 0.45, 1.15, 2.78 and 10.41 m from the water inlet.
- The film thickness was measured using the concentric probe type of conductance probe (used previously by Azzopardi (1986) & Chan (1990) for film thickness measurements). The probe output was sampled for 30 s at a frequency of 1 kHz. Wave velocity was measured using two conductance probes separated by an axial distance of 0.11 m. Wave frequency, defined here as that frequency at which the power is maximum, was obtained by taking the power spectrum of the film thickness signal after removing the DC component. Film thickness measurements were made at a distance of 0.05, 0.25, 0.94, 3.85 and 10.41 m from the water inlet.
- The wall shear stress was measured using a hot film probe developed by Govan, Owen, Hewitt, and Burnett (1989). It consists of a thin layer of nickel sputtered on to a polyamide foil. By means of photo etching process, the unwanted nickel film is removed and the probe is protected from electrolytic attack by a further layer of polyamide. A set of five probes, each 0.01 mm wide and 3 mm long and a resistance lying between 17 and 25 ohms, is glued onto a perspex plug having the same curvature as the pipe so that it is flush with the wall. The wires of the probes are fed through a pair of holes in the plug which are sealed with silicone. To avoid overheating, the hot film probe was operated at an overheat ratio of 0.5%. In the experiments, the shear stress probes were located at distances of 0.5, 1.15, 3.85 and 10.41 m from the water inlet. The probes were calibrated in situ by measuring the pressure gradient in single-phase flow.

In the experiments, the inlet pressure was fixed at 2.38 bar and the inlet temperature at 25°C. The liquid mass flux was varied between 10 and 120 kg/m² s (covering a liquid Reynolds number, based on the superficial velocity and the tube diameter, range of 300–3800) while maintaining the air flow rate constant at 71, 97, 124 and 154 kg/m² s, thus covering the rather high air flow Reynolds number range of 125,000–275,000. Pressure measurements were made during the tests at distances of 0.67, 1.59, 3.2 and 9.74 m downstream of the water inlet. For the highest flow rates, the pressure decreased by about 30% of its inlet value over the test section. For lower flow rates, the change was less. This change of pressure and the consequent change in the gas density was taken into account in calculating the various parameters of the flow.

3. Experimental results

From the experiments described in the above section, the following parameters of annular flow were obtained:

pressure gradient, film flow rate, time-averaged film thickness, time-averaged wall shear stress, disturbance wave characteristics, namely, their velocity and frequency. The variation of these parameters with distance from the water inlet gives some idea of the flow development and is discussed below. All the data are also given in a tabulated form in the Appendix.

3.1. Pressure gradient

The value measured here is the overall pressure gradient and includes the frictional, the accelerational and the gravitational components. However, it is expected that the frictional component would be the major contributor to the pressure gradient in annular flow. The variation of the pressure gradient with distance is summarized in Fig. 1. In Fig. 1a, the dimensionless pressure gradient measured at a gas flux of 71 kg/m² s is plotted at axial distances of 21, 50, 101 and 306 tube diameters (z/D) for various water mass fluxes. For the sake of clarity, the points belonging to the same air and water mass fluxes

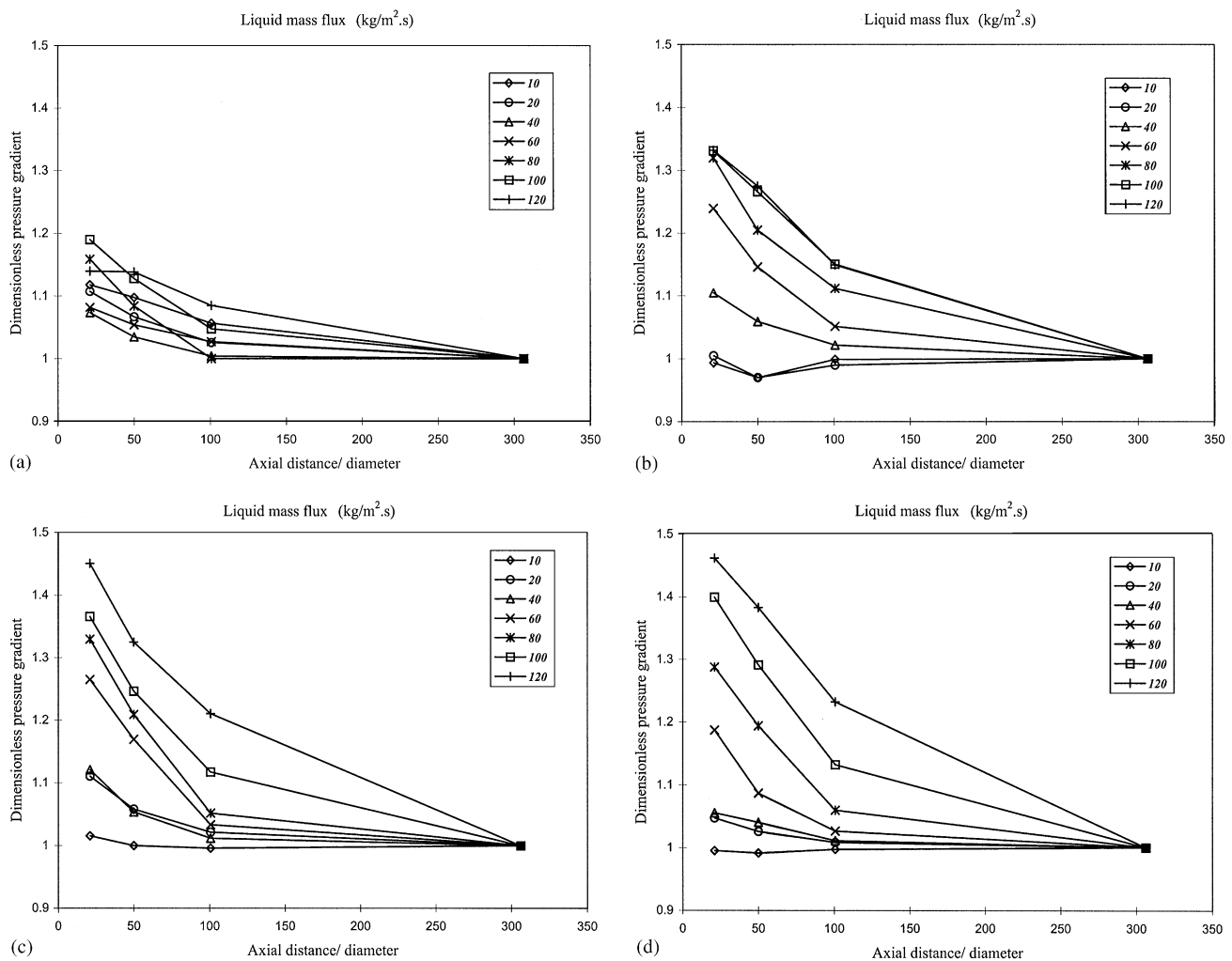


Fig. 1. Variation of dimensionless pressure gradient with axial distance for an air mass flux of (a) 71, (b) 97, (c) 124 and (d) 154 kg/m² s at different liquid mass fluxes.

are joined by straight lines. Figs. 1b–d show the same plot at air mass fluxes of 97, 124 & 154 kg/m² s, respectively. In these plots, the pressure gradient is rendered dimensionless by dividing its value at any axial location by its value at the last axial position measured at the same air and water flow rates. (Otherwise, the variation of the pressure gradient with air/water flow rate would mask its variation with distance at the same water/air flow rate.) Thus, all the curves end at a dimensionless pressure gradient value of 1 at a z/D of 306.

These plots bring out the effect of the air mass flux and the water mass flux on the flow development as far as the pressure gradient is concerned. It can be seen in Fig. 1a that the pressure gradient value close to the inlet is only about 10–20% higher than the value at 300 tube diameters. The effect of increasing the water mass flux is to increase the initial value. An increase in the air mass flux (Figs. 1b–d) accentuates this difference between the initial value and the value furthest downstream. Thus, at the highest air and water mass flux used in the experiment

(Fig. 1d), the pressure gradient value at 21 tube diameters is nearly 45% higher than its value at 306 tube diameters. Although it decreases rapidly with distance, its value at a z/D of 101 is still more than 20% higher. This indicates that although the pressure gradient reaches a steady value within the first 100 diameters or so for low air fluxes, it is still changing even after 300 diameters at high air and water flow rates. In all these experiments, the inlet pressure was kept constant at 2.38 bar. The higher pressure gradient at high flow rates may lead to a significant change in the air density and this may be the reason why the pressure gradient does not attain a steady value. Brown et al. (1975) reported the same observation in their experiments in which measurements were made up to nearly 600 tube diameters.

3.2. Wall shear stress

The variation of the measured wall shear stress with dimensionless axial distance (z/D) is shown in Fig. 2. As

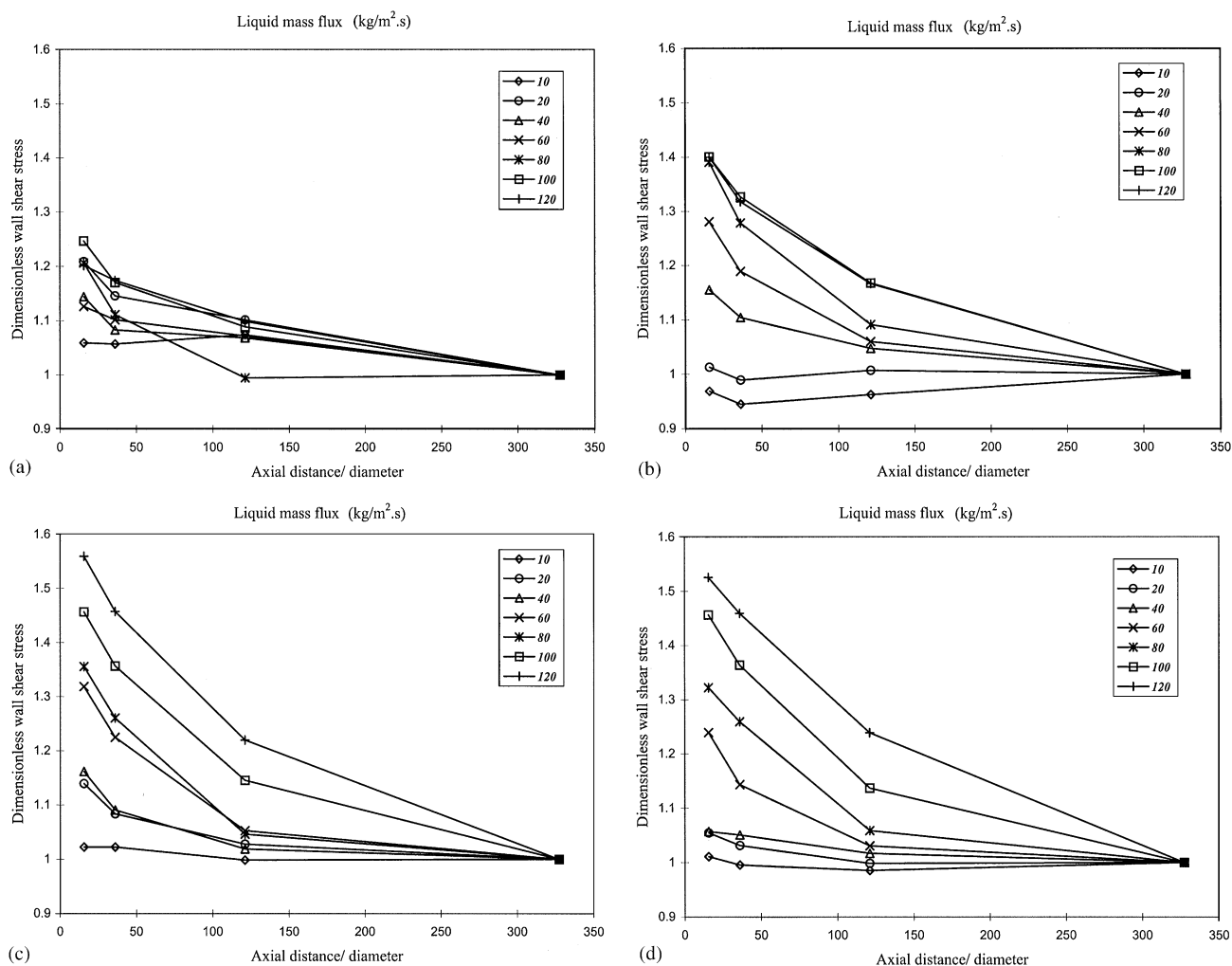


Fig. 2. Variation of dimensionless wall shear stress with axial distance for an air mass flux of (a) 71, (b) 97, (c) 124 and (d) 154 kg/m² s at different liquid mass fluxes.

before, Figs. 2a–d correspond to data obtained at a constant air mass flux of 71, 97, 124 and 154 kg/m² s, respectively. As has been done for the pressure gradient, the wall shear stress is made dimensionless by dividing it by its value at the last axial position measured at the same air and water flow rates.

An examination of Figs. 1 and 2 shows a striking similarity between the two, both qualitatively and quantitatively. This may justify two commonly made assumptions of annular flow, namely, that the wall shear stress is almost equal to the interfacial shear stress and that the pressure gradient in annular flow is dominated by interfacial friction, the accelerational and gravitational components being negligible. Under these conditions, one may write that

$$(\Delta P/\Delta z)_{\text{total}} \approx (\Delta P/\Delta z)_{\text{fric}} = 4(L/D)\tau_i \approx 4(L/D)\tau_w$$

$$\text{or } \Delta P \propto \tau_w,$$

which is evident from the present experimental results.

3.3. Film flow rate

The variation of the film flow rate with dimensionless axial distance (z/D) is shown in Fig. 3. As before, Figs. 3a–d correspond to data obtained at a constant air mass flux of 71, 97, 124 & 154 kg/m² s, respectively. In these plots, the water film flow rate is made dimensionless by dividing it by the inlet water flow rate. The subsequent film flow rate decreases due to entrainment, if any, and this is why the dimensionless film flow rate decreases with distance.

For the lowest water mass flux of 10 kg/m² s, there is no entrainment and hence the dimensionless film flow rate remains constant in all the plots. For other flow rates, it decreases rapidly with z/D ; the decrease being more rapid as the water/air flow rate increases. Thus, at the highest air and water mass flux (Fig. 3d), the dimensionless film flow rate drops to less than 0.30. There is one significant difference between the variation of film flow rate and that of the pressure gradient. In the case of the

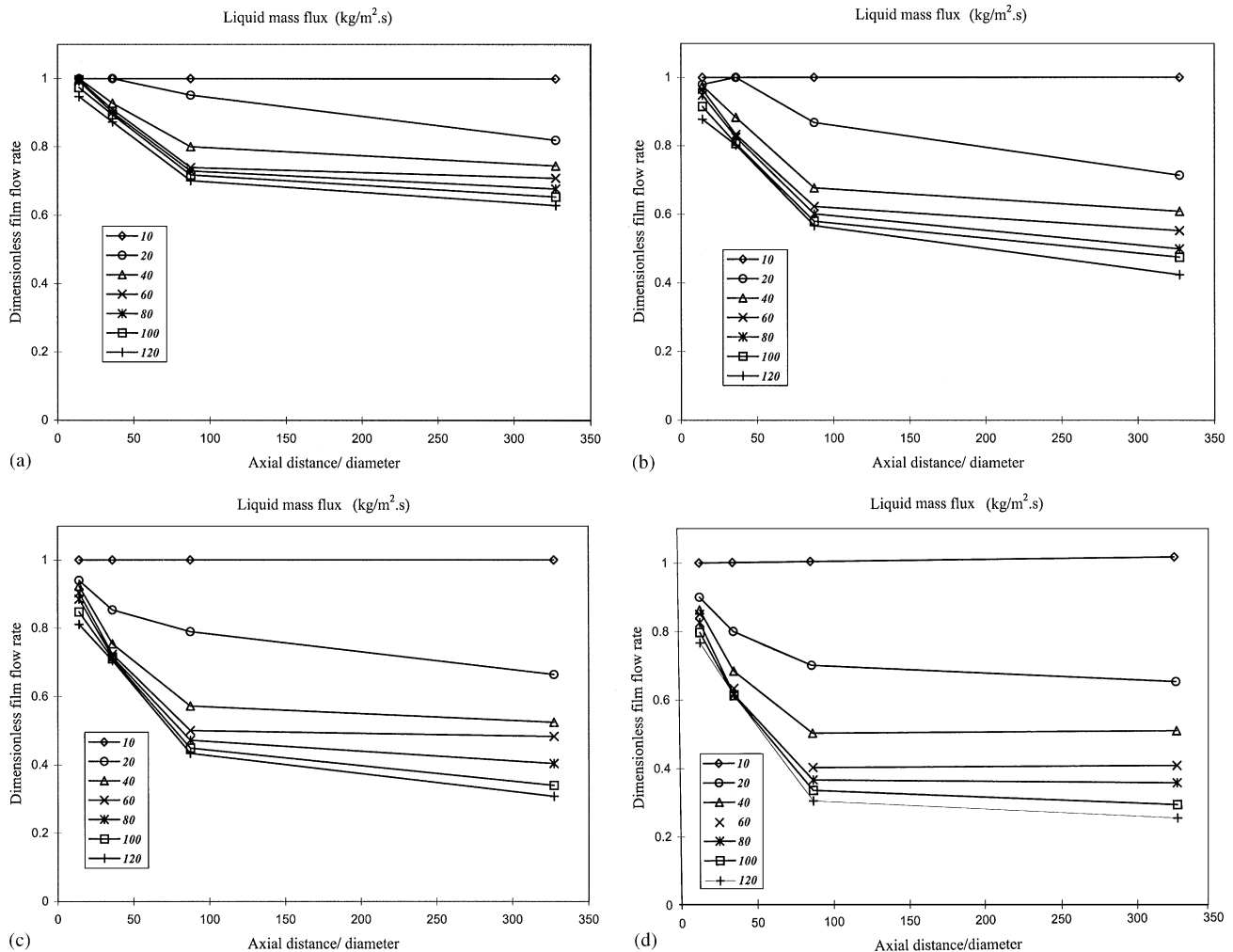


Fig. 3. Variation of liquid film flow rate with axial distance for an air mass flux of (a) 71, (b) 97, (c) 124 and (d) 154 kg/m² s at different liquid mass fluxes.

latter, the variation is still continuing even after 300 tube diameters whereas the dimensionless film flow rate shows much less variation between 100 and 300 tube diameters. This means that hydrodynamic equilibrium of the film flow (where deposition rate equals entrainment rate) is nearly established within the first 100 tube diameters at the highest air flow rate used in the present experiments. However, at lower air fluxes (Figs. 1a–c), significant difference in the film flow rate persists between 100 and 300 tube diameters, thus indicating that the flow is not yet fully developed.

3.4. Time-averaged film thickness

The variation of the time-averaged film thickness with dimensionless axial distance (z/D) is shown in Fig. 4. As before, Figs. 4a–d correspond to the data obtained at a constant air mass flux of 71, 97, 124 & 154 $\text{kg/m}^2 \text{ s}$, respectively. In these plots, the film thickness at any

location is made dimensionless by dividing it by its value at the last axial position for the same air and water fluxes.

It can be seen that the film thickness varies over a wider range than do the other parameters and also shows some anomalous behaviour. Close to the inlet, its value seems to increase slowly with distance followed by a rapid decrease. This is rather unexpected because one would expect the film thickness to decrease as the film flow rate would decrease due to entrainment. This surprising result may be explained as follows. Disturbance waves are not formed at the inlet but take a finite distance to develop. Power spectra of the film thickness measured at various distances from the inlet (Fig. 5) show that disturbance waves are formed after about 10 tube diameters from the inlet. Thus, until this point, there cannot be entrainment of the liquid and the film thickness therefore cannot decrease due to entrainment. This is also consistent with the fact that in Fig. 4, the film thickness increases up to 8 diameters in almost all cases

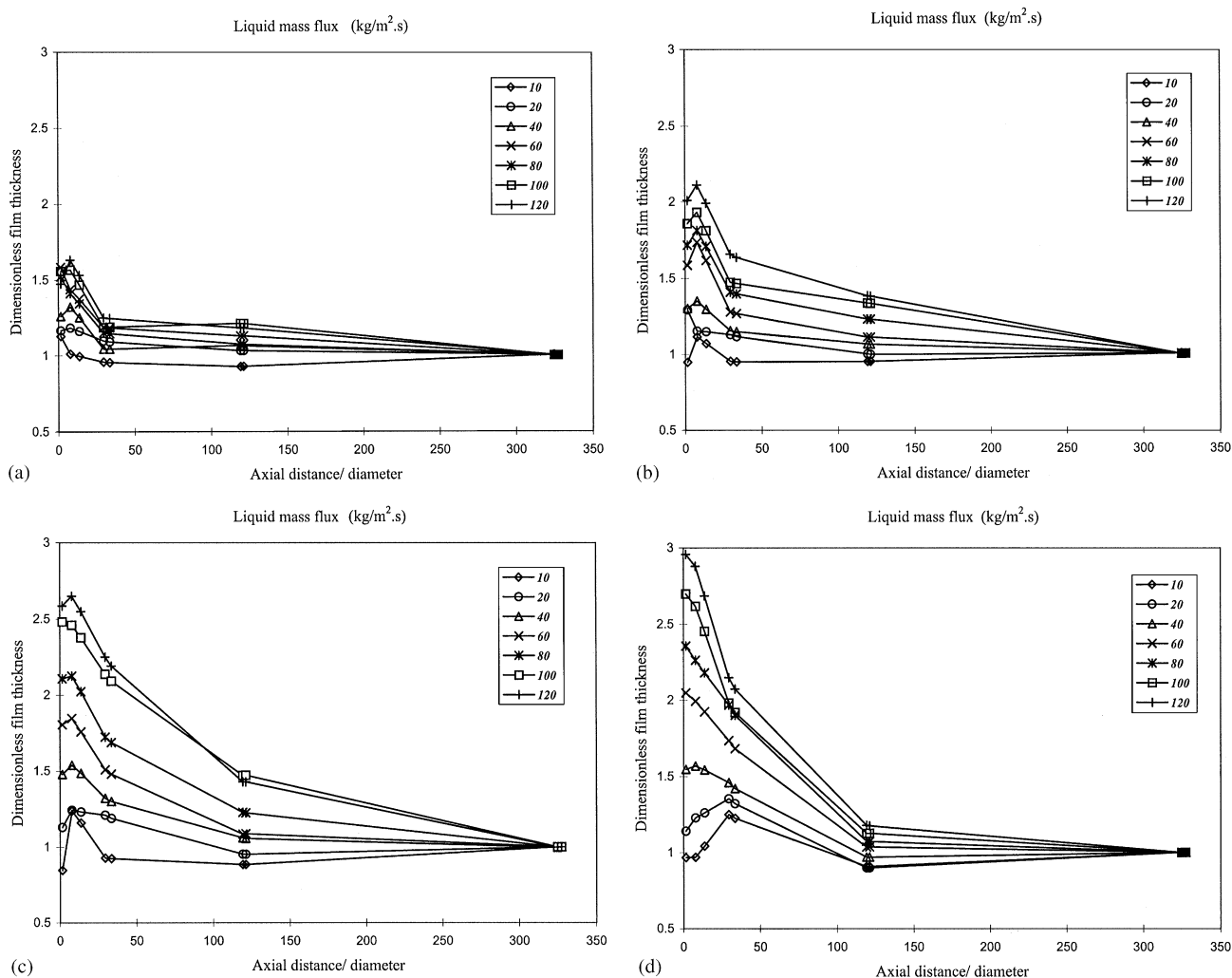


Fig. 4. Variation of liquid film thickness with axial distance for an air mass flux of (a) 71, (b) 97, (c) 124 and (d) 154 $\text{kg/m}^2 \text{ s}$ at different liquid mass fluxes.

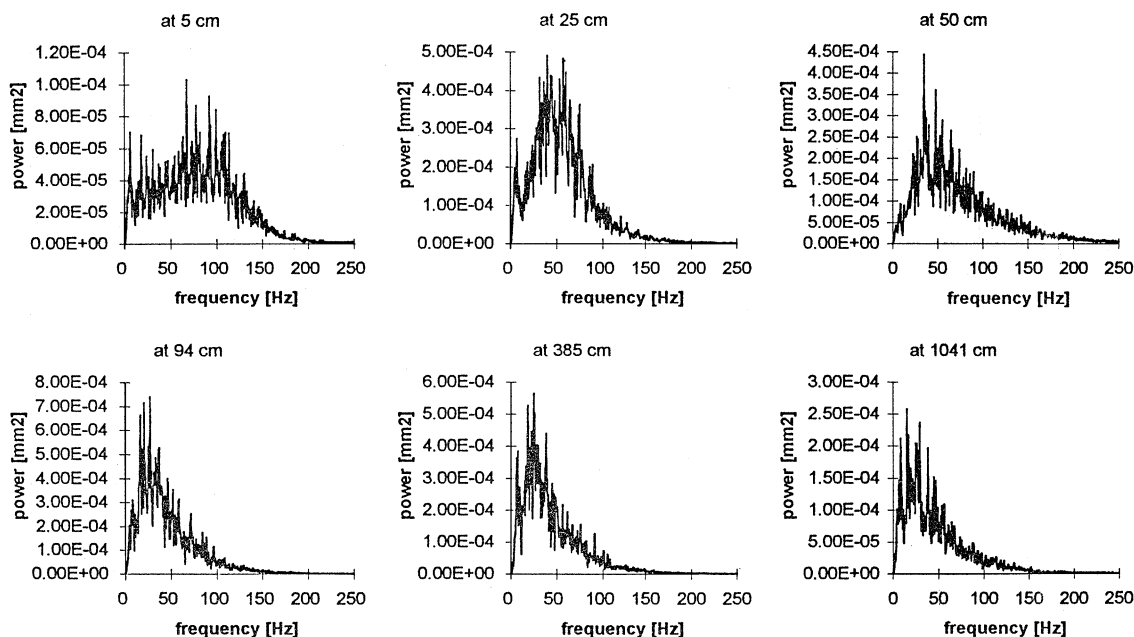


Fig. 5. Power spectra of film thickness variation for an air flow rate of $71 \text{ kg/m}^2 \text{ s}$ and a water flow rate of $20 \text{ kg/m}^2 \text{ s}$ at distances of 0.05, 0.25, 0.5, 0.94, 3.85 and 10.41 m from the water inlet.

and then starts decreasing by the next measuring station, which was at 15 diameters from the inlet by which point disturbance waves may have been formed and entrainment may have started. The actual increase of the liquid film very close to the inlet may be due to the transition between laminar and turbulent flow, the liquid film in turbulent flow being thicker. Another noteworthy feature of the film thickness is that it varies more strongly with distance, especially at high air flow rates (Fig. 4d). As far as the development length is concerned, the film thickness does not change much between 120 and 330 diameters for the highest air flow rate used in the experiments; at lower air flow rates this change is appreciable. In this respect, the film thickness shows a trend similar to that of the film flow rate (Fig. 3).

3.5. Interfacial wave spectrum

The film thickness variation close to the water inlet shows an interesting pattern. The typical evolution with distance of the interfacial wave spectrum is illustrated in Fig. 5. Here the autospectral density function of the film thickness trace (after removing its DC component) is shown for an air flow rate of $71 \text{ kg/m}^2 \text{ s}$ and a water flow rate of $20 \text{ kg/m}^2 \text{ s}$ at distances of 0.05, 0.25, 0.5, 0.94, 3.85 and 10.41 m from the water inlet. In Fig. 5a, i.e., at a distance of only one and a half tube diameters from the water inlet, there is essentially no disturbance wave pattern and the dominant wave frequency in the spectrum occurs at around 70 Hz. In Fig. 5b, i.e., at a distance of 8 diameters, a disturbance wave-like pattern is emerging

in the sense that (i) the dominant frequency is about 30 Hz, and (ii) the magnitude of the maximum power (i.e., the square of the film thickness variation) has increased five-fold over that in Fig. 5a. Further downstream, the range of dominant frequencies narrows down and the disturbance wave frequency also decreases. This decrease has earlier been attributed to wave coalescence (Hall-Taylor & Nedderman, 1968) and their model was found to work satisfactorily for the present data (see below).

The probability density function of the film thickness corresponding to the above cases is given in Fig. 6. Very close to the inlet (Fig. 6a), the film is fairly uniform and does not show large fluctuations. At eight diameters from the water inlet (Fig. 6b), the process of formation of large waves is becoming evident because the maximum film height *increases* from about 0.35 mm in Fig. 5a to 0.55 mm in Fig. 6b. At the same time, the probability density function begins to exhibit a well-defined peak corresponding to the thickness of the sublayer between two successive disturbance waves (Jayanti & Hewitt, 1991). This process of formation of disturbance waves continues with further distance (Fig. 6c) and the probability value of the peak increases with distance (Figs. 6d and e) showing that the separation distance between successive disturbance waves is increased, presumably due to the coalescence of disturbance waves (Hall-Taylor & Nedderman, 1968).

Thus, it appears that the interfacial structure takes a fairly short distance, of the order of 50–100 diameters, to develop completely. This is in agreement with the

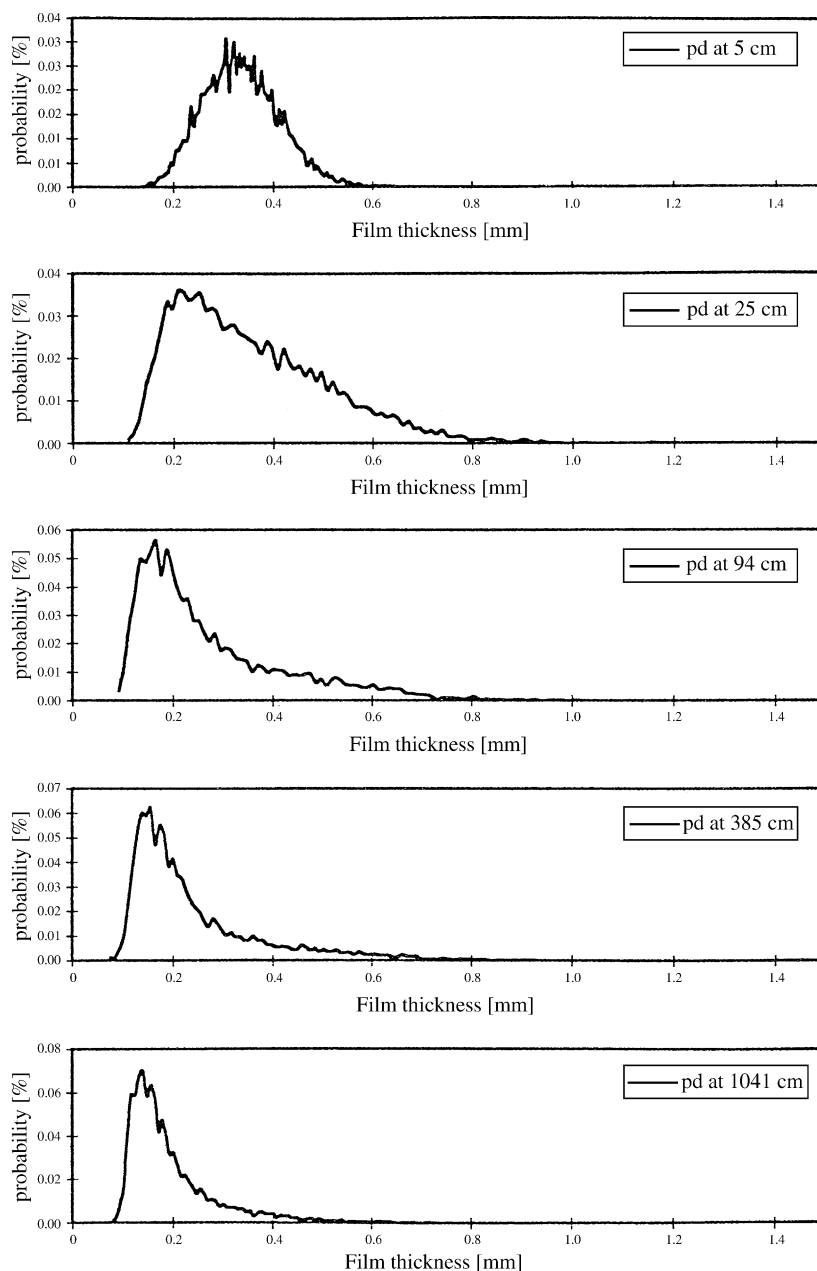


Fig. 6. Probability density function of the film thickness for the cases shown in Fig. 5.

observation of Brown et al. (1975) based on their steam/water data.

3.6. Disturbance wave frequency and velocity

The disturbance wave frequency can be obtained by evaluating the autospectral density function of the film thickness trace. The dominant frequency here corresponds to the frequency of the disturbance waves. Their velocity can be determined by cross-correlating the film thickness traces measured simultaneously using probes set apart by a known distance.

The measured variation of the disturbance wave frequency with distance is shown in Fig. 7 in terms of dimensionless parameters. Here, the wave frequency at any location is made dimensionless by dividing it by its value at the last axial position for the same flow conditions. In all cases, the disturbance wave frequency decreases with distance. Unlike in the case of film flow rate and film thickness, this relative wave frequency does not appear to be a function of the water mass flux except for the lowest water mass flux ($10 \text{ kg/m}^2 \text{ s}$) at which point disturbance waves could not be seen clearly and the results are therefore probably misleading or erroneous.

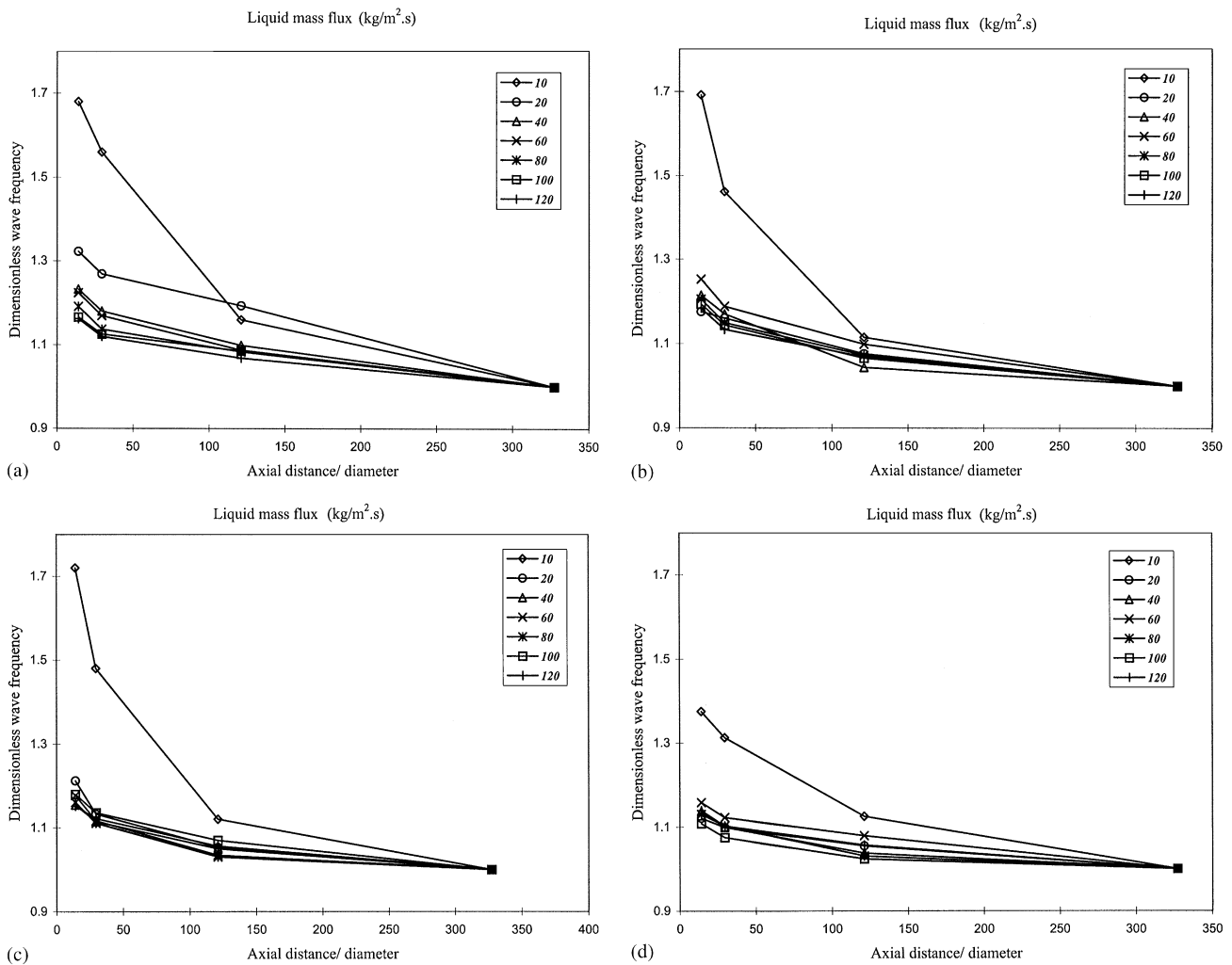


Fig. 7. Variation of the non-dimensionalized disturbance wave frequency with axial distance for an air mass flux of (a) 71, (b) 97, (c) 124 and (d) 154 kg/m² s at different liquid mass fluxes.

This relative frequency decreases with increasing air mass flux (see Figs. 7a–d) which may be due to the increasing frequency of the waves itself.

Hall-Taylor and Nedderman (1968) proposed that this decrease of wave frequency with distance was due to wave coalescence and developed an empirical correlation to predict this rate of change. The correlation was based on the idea that a distribution of wave velocities near the point of formation of disturbance waves would lead to coalescence and hence a reduction in the frequency of the waves downstream. If the initial wave frequency and the standard deviation of the average wave velocity are known, this correlation can be used to calculate the frequency at any distance. In the present experiments, the initial frequency, i.e., at the point of origin, is not known. Taking the standard deviation of the disturbance wave velocity to be the same as that of Hall-Taylor and Nedderman, the initial frequency can be back-calculated from the measured frequency at a particular axial loca-

tion. It was found that the initial frequency thus calculated was nearly the same for all downstream distances for a given air and water flow rate combination. Also, this calculated initial frequency was found to be in good agreement with the turbulent burst frequency in an equivalent liquid film flow (Martin & Azzopardi, 1985). While these calculations are not a direct evidence for either the wave coalescence mechanism or the relation between disturbance waves and turbulence bursts, the results do provide circumstantial evidence to support these hypotheses.

The variation of the measured disturbance wave velocity with distance from the water inlet is shown in Fig. 8. It is seen that the wave velocity increases with distance initially, but later becomes fairly constant. Here, there are two opposing factors which may affect the wave velocity: decreasing film thickness (due to entrainment) which tends to reduce the wave velocity and increasing air velocity (due to decreasing pressure) which tends to

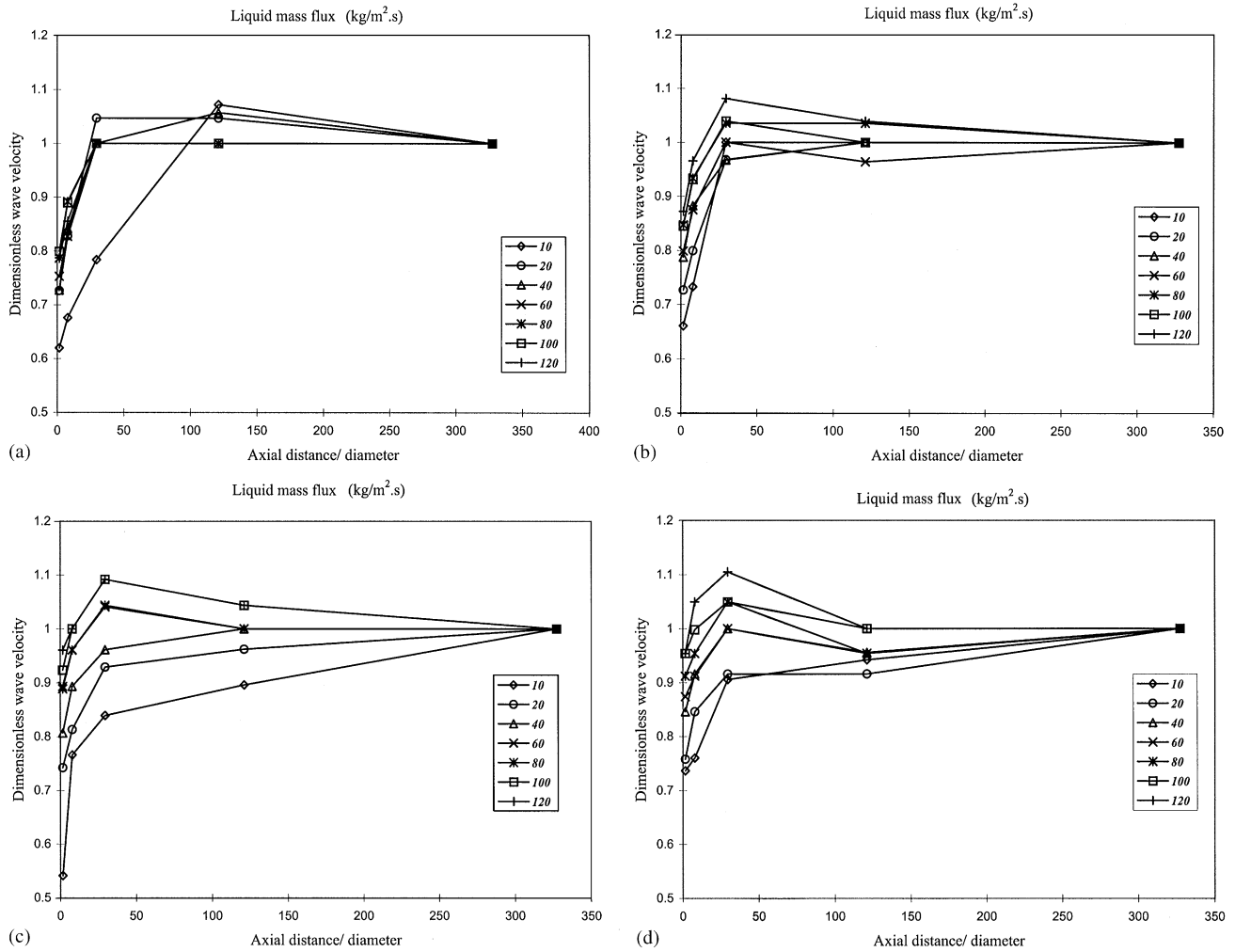


Fig. 8. Variation of non-dimensionalized disturbance wave velocity with axial distance for an air mass flux of (a) 71, (b) 97, (c) 124 and (d) 154 kg/m² s at different liquid mass fluxes.

increase the wave velocity. It appears that the two are nearly cancelling the effect of each other with the net result that there is little overall variation of the wave velocity with distance.

3.8. Discussion

The results from the experiments show that annular flow takes considerable distance to develop. While all the important flow parameters show appreciable variation in the region close to the inlet, they reach a quasi-steady state at different distances from the inlet. The parameters related to the interfacial wave structure, namely, the wave spectrum, disturbance wave velocity and frequency, showed only a small variation between z/D of 100 and 300 for all flow rate combinations of air and water. Film flow rate and mean film thickness took a longer distance to develop and showed appreciable variation between 100 and 300 diameters for high liquid flow rates. For the highest air mass flux used in the present study, these

parameters showed little change after 100 diameters. The pressure gradient and the wall shear stress continue to change even after 100 pipe diameters even for the highest air flow rate condition. This is most likely due to the large pressure drop which occurs under these conditions as a result of which the gas continues to accelerate.

It may therefore be concluded that for a flow where the liquid phase is introduced through a porous wall, the flow may take a fairly short distance, of the order of 100 diameters, to develop at high air flow rates. Any changes that continue to occur further are mostly due to the change in gas density due to pressure drop in the test section. Film flow rate and film thickness are the slowest to respond and these may take a much longer distance, 100–300 diameters, to develop fully.

Appendix

The data of pressure gradient, wall shear stress, film thickness, film flow rate, wave frequency and wave

Table 1
Local test section pressure (absolute) at various distances from the liquid inlet

Air mass flux (kg/m ² s)	Water mass flux (kg/m ² s)	Local test section pressure (in bar) at a distance from the liquid inlet of						
		0 m	0.51 m	1.02 m	1.61 m	2.72 m	3.82 m	10.01 m
71	10	2.379	2.363	2.347	2.329	2.295	2.261	2.069
71	20	2.379	2.356	2.334	2.309	2.263	2.218	1.965
71	40	2.379	2.350	2.324	2.294	2.241	2.188	1.896
71	60	2.379	2.346	2.315	2.281	2.220	2.160	1.827
71	80	2.379	2.344	2.311	2.275	2.210	2.146	1.793
71	100	2.379	2.342	2.308	2.270	2.200	2.132	1.758
71	120	2.379	2.341	2.305	2.270	2.190	2.118	1.724
97	10	2.379	2.353	2.326	2.296	2.239	2.182	1.862
97	20	2.379	2.349	2.320	2.287	2.226	2.166	1.827
97	40	2.379	2.342	2.308	2.270	2.202	2.134	1.758
97	60	2.379	2.337	2.298	2.255	2.179	2.105	1.690
97	80	2.379	2.335	2.294	2.249	2.169	2.090	1.655
97	100	2.379	2.330	2.286	2.237	2.148	2.061	1.586
97	120	2.379	2.329	2.283	2.231	2.138	2.047	1.552
124	10	2.379	2.349	2.319	2.285	2.220	2.156	1.793
124	20	2.379	2.340	2.304	2.265	2.194	2.123	1.724
124	40	2.379	2.336	2.296	2.252	2.172	2.094	1.655
124	60	2.379	2.330	2.284	2.235	2.147	2.062	1.586
124	80	2.379	2.325	2.276	2.222	2.126	2.032	1.517
124	100	2.379	2.321	2.267	2.209	2.104	2.003	1.448
124	120	2.379	2.319	2.263	2.202	2.092	1.987	1.414
154	10	2.379	2.346	2.312	2.274	2.201	2.129	1.724
154	20	2.379	2.339	2.302	2.260	2.184	2.107	1.676
154	40	2.379	2.332	2.288	2.240	2.153	2.067	1.586
154	60	2.379	2.325	2.276	2.222	2.127	2.035	1.517
154	80	2.379	2.319	2.264	2.205	2.102	2.002	1.448
154	100	2.379	2.314	2.254	2.190	2.078	1.970	1.379
154	120	2.379	2.312	2.251	2.184	2.067	1.955	1.345

Table 2
Measured pressure gradient (in Pa/m) at various distances from the liquid inlet

Air mass flux (kg/m ² s)	Water mass flux (kg/m ² s)	Pressure gradient (in Pa/m) at a distance (of the mid-point of the pressure tappings) from the liquid inlet of			
		0.67 m	1.59 m	3.20 m	9.74 m
71	10	1220	1198	1153	1092
71	20	1560	1503	1445	1409
71	40	2075	2000	1941	1934
71	60	2713	2643	2576	2508
71	80	3196	2990	2759	2759
71	100	3489	3307	3071	2932
71	120	3694	3690	3517	3241
97	10	2006	1958	2017	2019
97	20	2401	2318	2366	2390
97	40	3039	2914	2810	2751
97	60	4025	3722	3415	3248
97	80	4675	4268	2928	3542
97	100	5033	4787	4348	3780
97	120	5539	5304	4779	4159
124	10	2988	2942	2930	2942
124	20	3802	3624	3497	3425
124	40	4518	4248	4079	4032
124	60	5708	5277	4663	4513
124	80	6694	6086	5295	5035
124	100	7239	6605	5920	5300
124	120	7994	7302	6671	5511
154	10	4729	4711	4739	4752
154	20	5617	5503	5408	5364
154	40	6169	6081	5910	5845
154	60	7796	7138	6741	6568
154	80	8904	8255	7327	6915
154	100	9889	9124	8001	7067
154	120	10518	9950	8863	7195

Table 3
Measured wall shear stress (in Pa) at various distances from the liquid inlet

Air mass flux (kg/m ² s)	Water mass flux (kg/m ² s)	Wall shear stress (in Pa) at a distance from the liquid inlet of			
		0.50 m	1.15 m	3.85 m	10.41 m
71	10	10.75	10.73	10.90	10.15
71	20	14.77	14.00	13.46	12.22
71	40	19.61	18.56	18.31	17.14
71	60	24.73	24.19	23.54	21.96
71	80	30.44	28.04	25.10	25.24
71	100	32.50	30.51	28.38	26.07
71	120	34.83	34.04	31.85	28.99
97	10	16.95	16.53	16.84	17.50
97	20	20.60	20.12	20.48	20.34
97	40	26.61	25.44	24.13	23.04
97	60	34.81	32.32	28.81	27.18
97	80	41.70	38.34	32.72	29.99
97	100	44.13	41.81	36.79	31.51
97	120	48.45	45.61	40.37	34.61
124	10	24.51	24.51	23.94	23.98
124	20	31.83	30.27	28.71	27.94
124	40	38.51	36.15	33.77	33.15
124	60	48.08	44.65	38.37	36.46
124	80	55.92	51.99	43.14	41.25
124	100	61.65	57.42	48.47	42.32
124	120	68.38	63.89	53.47	43.84
154	10	38.55	37.97	37.58	38.14
154	20	45.75	44.74	43.31	43.38
154	40	49.79	49.49	47.90	47.10
154	60	64.80	59.78	53.91	52.30
154	80	73.31	69.82	58.71	55.45
154	100	81.99	76.76	63.98	56.29
154	120	86.84	83.07	70.50	56.92

Table 4
Measured liquid film mass flux (in kg/m² s) at various distances from liquid inlet

Air mass flux (kg/m ² s)	Water mass flux (kg/m ² s)	Liquid film flux (in kg/m ² s) at a distance from the liquid inlet of			
		0.45 m	1.15 m	2.78 m	10.41 m
71	10	10.0	10.0	10.0	10.0
71	20	20.0	20.0	19.0	16.4
71	40	40.0	37.1	32.0	29.8
71	60	59.8	54.6	44.4	42.5
71	80	79.4	72.0	58.3	54.2
71	100	97.4	89.4	71.8	65.5
71	120	113.7	104.7	84.1	75.6
97	10	10.0	10.0	10.0	10.0
97	20	19.6	20.0	17.4	14.3
97	40	39.0	35.3	27.1	24.3
97	60	57.9	49.9	37.4	33.1
97	80	75.8	66.0	48.1	39.9
97	100	91.5	80.6	57.9	47.4
97	120	105.3	96.2	68.0	50.8
124	10	10.0	10.0	10.0	10.0
124	20	18.8	17.1	15.8	13.3
124	40	36.9	30.2	22.9	21.0
124	60	54.2	43.4	30.0	29.0
124	80	70.8	57.3	37.8	32.3
124	100	84.8	71.1	44.9	34.0
124	120	97.4	84.6	52.0	36.9
154	10	10.0	10.0	10.0	10.0
154	20	18.0	16.0	13.9	12.7
154	40	34.5	27.3	19.9	19.7
154	60	51.0	37.9	23.9	23.5
154	80	66.1	49.1	29.0	27.3
154	100	79.7	61.1	33.2	27.7
154	120	92.0	73.5	36.1	28.5

Table 5
Measured film thickness (in mm) at various distances from the liquid inlet

Air mass flux (kg/m ² s)	Water mass flux (kg/m ² s)	Mean film thickness (mm) at a distance from the liquid inlet of						
		0.05 m	0.25 m	0.44 m	0.94 m	1.07 m	3.85 m	10.4 m
71	10	0.164	0.147	0.145	0.140	0.139	0.135	0.146
71	20	0.199	0.202	0.198	0.187	0.186	0.176	0.171
71	40	0.291	0.305	0.289	0.241	0.241	0.246	0.231
71	60	0.431	0.391	0.374	0.323	0.322	0.307	0.272
71	80	0.513	0.476	0.453	0.388	0.386	0.362	0.338
71	100	0.546	0.549	0.514	0.415	0.416	0.424	0.350
71	120	0.551	0.610	0.572	0.467	0.465	0.441	0.374
97	10	0.110	0.129	0.124	0.111	0.110	0.110	0.116
97	20	0.156	0.140	0.139	0.137	0.135	0.121	0.121
97	40	0.203	0.210	0.202	0.180	0.179	0.166	0.156
97	60	0.273	0.298	0.278	0.220	0.218	0.191	0.172
97	80	0.337	0.355	0.335	0.276	0.274	0.241	0.196
97	100	0.382	0.397	0.372	0.303	0.301	0.274	0.205
97	120	0.419	0.440	0.415	0.346	0.341	0.288	0.208
124	10	0.076	0.111	0.104	0.083	0.083	0.079	0.090
124	20	0.107	0.118	0.117	0.115	0.113	0.091	0.095
124	40	0.160	0.167	0.161	0.143	0.141	0.115	0.108
124	60	0.215	0.220	0.209	0.179	0.176	0.129	0.119
124	80	0.257	0.260	0.247	0.210	0.206	0.150	0.122
124	100	0.295	0.293	0.283	0.254	0.249	0.175	0.119
124	120	0.319	0.327	0.314	0.277	0.270	0.176	0.123
154	10	0.070	0.070	0.076	0.090	0.089	0.066	0.072
154	20	0.088	0.095	0.097	0.105	0.102	0.069	0.077
154	40	0.126	0.128	0.126	0.119	0.116	0.079	0.082
154	60	0.169	0.165	0.159	0.143	0.139	0.086	0.083
154	80	0.203	0.195	0.188	0.170	0.164	0.093	0.086
154	100	0.238	0.231	0.216	0.175	0.169	0.099	0.088
154	120	0.261	0.254	0.237	0.190	0.183	0.104	0.088

Table 6
Measured disturbance wave frequency (in Hz) at various distances from liquid inlet

Air mass flux (kg/m ² s)	Water mass flux (kg/m ² s)	Disturbance wave frequency (in Hz) at a distance from the liquid inlet of			
		0.45 m	0.94 m	3.85 m	10.41 m
71	10	4.2	3.9	2.9	2.5
71	20	12.3	11.8	11.1	9.3
71	40	21.2	20.3	18.9	17.2
71	60	22.4	21.4	19.9	18.3
71	80	24.3	23.2	22.1	20.4
71	100	26.0	25.1	24.2	22.3
71	120	27.2	26.2	25.0	23.4
97	10	4.4	3.8	2.9	2.6
97	20	15.4	15.2	14.1	13.1
97	40	22.1	21.3	19.0	18.2
97	60	25.3	24.0	22.2	20.2
97	80	28.2	26.9	25.1	23.4
97	100	29.0	27.8	25.9	24.3
97	120	31.0	29.7	28.0	26.2
124	10	4.3	3.7	2.8	2.5
124	20	18.3	17.1	15.9	15.1
124	40	23.2	22.4	21.1	20.1
124	60	27.3	25.9	24.0	23.2
124	80	30.4	29.1	27.0	26.2
124	100	32.2	31.0	29.2	27.3
124	120	35.0	34.1	32.1	30.4
154	10	4.4	4.2	3.6	3.2
154	20	20.4	20.0	19.2	18.2
154	40	24.0	23.2	21.9	21.1
154	60	29.3	28.4	27.3	25.3
154	80	33.0	32.2	30.1	29.2
154	100	36.9	35.8	34.1	33.3
154	120	40.1	38.8	37.2	35.2

Table 7

Measured disturbance wave velocity (in m/s) at various distances from liquid inlet

Air mass flux (kg/m ² s)	Water mass flux (kg/m ² s)	Disturbance wave velocity (in m/s) at a distance from the liquid inlet of			
		0.50 m	1.15 m	3.85 m	10.41 m
71	10	1.77	2.00	2.68	2.50
71	20	2.29	2.62	2.68	2.56
71	40	2.68	2.97	3.14	2.97
71	60	2.89	3.24	3.24	3.24
71	80	3.06	3.33	3.33	3.33
71	100	3.14	3.44	3.44	3.44
71	120	3.33	3.67	3.67	3.67
97	10	2.68	3.33	3.33	3.33
97	20	2.97	3.33	3.44	3.44
97	40	3.33	3.55	3.67	3.67
97	60	3.55	3.93	3.79	3.93
97	80	3.79	4.07	4.07	3.93
97	100	3.93	4.23	4.07	4.07
97	120	4.07	4.40	4.23	4.07
124	10	3.33	3.55	3.79	4.23
124	20	3.55	3.93	4.07	4.23
124	40	4.07	4.23	4.40	4.40
124	60	4.40	4.58	4.40	4.40
124	80	4.58	4.78	4.58	4.58
124	100	4.78	5.00	4.78	4.58
124	120	4.78	5.00	4.78	4.58
154	10	3.79	4.23	4.40	4.67
154	20	4.40	4.58	4.58	5.00
154	40	4.78	5.00	4.78	5.00
154	60	5.00	5.24	5.00	5.24
154	80	5.24	5.50	5.00	5.24
154	100	5.24	5.50	5.24	5.24
154	120	5.50	5.79	5.24	5.24

velocity obtained in the present study for different flow rates of air and water are tabulated below. In all the cases, the inside tube diameter should be taken as 0.0318 m. The pressure at the air inlet was kept constant; however, the local static pressure varied along the test section depending on the flow rates. The static pressure at various measuring stations is given in Table 1 for different air and water flow rates. The variation of the pressure gradient along the test section is given in Table 2 with the reported axial distance being the distance of the mid-point of the two tappings from the liquid inlet. The time-averaged axial wall shear stress is given in Table 3; the liquid film flow rate in Table 4; the mean film thickness in Table 5; the disturbance wave frequency in Table 6 and the disturbance wave velocity in Table 7. The last two quantities were inferred from a spectral analysis of the instantaneous film thickness measured at various axial locations.

References

- Asali, J., Leman, G., & Hanratty, T. J. (1985). Entrainment measurements and their use in design equations. *PhysicoChemical Hydrodynamics*, 6, 207–221.
- Azzopardi, B. J. (1986). Disturbance wave frequencies, velocities and spacing in vertical annular two-phase flow. *Nuclear Engineering Design*, 92, 121–133.
- Brown, D. J., Jensen, A., & Whalley, P. B. (1975). *Non-equilibrium effects in heated and unheated annular two-phase flow*. ASME Paper No. 75-WA/HT-7.
- Chan, G. T. W. C. (1990). *Evaporation and condensation in annular vertical upward flow of water and steam*. Ph.D. thesis, Imperial College, London, UK.
- Gill, L. E., Hewitt, G. F., & Roberts, D. N. (1969). *Studies of the behaviour of the disturbance waves in annular flow in a long vertical tube*. UKAEA report no. AERE- R6012.
- Govan, A. H. (1990). *Modelling of vertical annular and dispersed two-phase flows*. Ph.D. thesis, Imperial College, London, UK.
- Govan, A. H., Owen, D. G., Hewitt, G. F., & Burnett, G. (1989). Wall shear stress measurements in vertical air–water annular two-phase flow. *International Journal of Multiphase Flow*, 15, 307–325.
- Hall-Taylor, N. S. (1966). *Interfacial phenomena in vertical annular two-phase flow*. Ph.D. thesis, Trinity College, Cambridge University, UK.
- Hall-Taylor, N. S., & Nedderman, R. M. (1968). The coalescence of disturbance waves in annular flow. *Chemical Engineering Science*, 23, 551–564.
- Hewitt, G. F., & Hall-Taylor, N. S. (1970). *Annular two-phase flow*. Oxford, UK: Pergamon.

- Hewitt, G. F., & Nicholls, B. (1969). *Film thickness measurement in annular two-phase flow using a fluorescence spectrometer technique, Part II: Studies of the shape of the disturbance waves*. UKAEA report no. AERE-R 4506.
- Jayanti, S., & Hewitt, G. F. (1991). Structure of interfacial waves in air–water horizontal annular flow. In: G. F. Hewitt, F. Mayinger, & J. R. Riznic (Eds.), *Phase-interface phenomena in multiphase flow* (pp. 95–104). New York, USA: Hemisphere.
- Martin, C. J. (1983). *Annular two-phase flow*. D.Phil. thesis, Oxford University, UK.
- Martin, C. J., & Azzopardi, B. J. (1985). Waves in vertical annular flow. *PCH PhysicoChemical Hydrodynamics*, 6, 257–265.
- Owen, D. G., & Hewitt, G. F. (1987). *An improved annular two-phase flow model*. Paper presented at the Third International Symposium on Multiphase Flow, the Hague, Netherlands.
- Sekoguchi, K., & Mori, K. (1997). New developments of experimental study of interfacial structure in gas–liquid two-phase flow. *Experimental Heat Transfer, Fluid Mechanics and Thermodynamics*, 2, 1177–1188.
- Sekoguchi, K., & Takeishi, M. (1989). Interfacial structures in upward huge wave flow and annular flow regimes. *International Journal of Multiphase Flow*, 15, 295–305.
- Tomida, T., & Okazaki, T. (1974). Statistical character of large disturbance waves in upward two-phase flow of air–water mixtures. *Journal of the Chemical Engineering of Japan*, 7, 329.
- Whalley, P. B., Hewitt, G. F., Azzopardi, B. J., & Pshyk, L. (1977). *Axial view photography of waves in annular two-phase flow*. UKAEA report no. AERE-R 8787.
- Wilkes, N. S., Azzopardi, B. J., & Thompson, C. P. (1983). Wave coalescence and entrainment in vertical annular two-phase flow. *International Journal of Multiphase Flow*, 9, 383–398.
- Wolf, A., Jayanti, S., & Hewitt, G. F. (1996). On the nature of ephemeral waves in vertical annular flow. *International Journal of Multiphase Flow*, 22, 325–333.